Politecnico di Milano
Dipartimento di Elettronica, Informazione e Bioingegneria
prof. Fabrizio Ferrandi

Parallel Computing– II part–Thursday, June 8th, 2023

Polimi ID ________ ____________________________

Surname ________ ______________________ **Name** _________________________

- This is a closed-book examination. You cannot use computers, phones, or laptops during the exam.
- Paper will be provided, but you should bring and use writing instruments that yield marks dark enough to be read easily. Erasable pens can be used.
- Total available time: 1h:15m.

Exercise 1 (4 points) _________________________

Exercise 2 (4 points) _________________________

Exercise 3 (4 points) _________________________

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Exercise n. 1

Answer the following questions about parallel patterns and briefly explain (without an explanation, the answer will be considered invalid)

A. Please define what a parallel pattern stencil is: assumptions, complexity, and example. (1)

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B. Are there collision problems with the Gather pattern? (1)

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C. Please describe how the Unsplit pattern works. (1)

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D. Can you remove a flow dependency? (1)

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Exercise n. 2

Answer the following questions about PThreads and OpenMP (without an explanation, the answer will be considered invalid).

A. Describe briefly how mutex variables work in PThread. (1)

B. Describe briefly how condition variables work in PThread. (1)

C. What is the difference between static and dynamic schedule of OpenMP for loops? (1)

D. What happens when a variable is declared as firstprivate in an OpenMP pragma? (1)

Exercise n. 3

Answer the following questions about parallel programming models and heterogeneous computing (without an explanation, the answer will be considered invalid).

A. Compare MPI, Pthreads, OpenMP, and CUDA according to how they describe parallelism and communication (i.e., implicitly/explicitly). (2)

B. Why is heterogeneous processing becoming more and more relevant in HPC? (2)

Exercise n. 4

Answer the following questions about GPGPU computing with CUDA (without an explanation, the answer will be considered invalid).

A. What are thread hierarchy variables in CUDA and why are they useful? (1)

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B. How is error handling implemented in a CUDA host program to check whether a kernel launch has been successful? (1)

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C. What is the impact of control divergence for the following kernel if it is executed with blocks of 32x32 threads on a 1000x1000 input? Propose a different block size that results in less divergence. (2)

```
__global__ void PictureKernel(float* d_Pin, float* d_Pout, int n, int m) {
    // Calculate the row id of the d_Pin and d_Pout element to process
    int Row = blockIdx.y*blockDim.y + threadIdx.y;
    // Calculate the column id of the d_Pin and d_Pout element to process
    int Col = blockIdx.x*blockDim.x + threadIdx.x;
    // each thread computes one element of d_Pout if in range
    if ((Row < m) && (Col < n)) {
        d_Pout[Row*n+Col] = 2*d_Pin[Row*n+Col];
    }
}
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